THE GOLDILOCKS THRESHOLD: QUANTITATIVE LIMITS OF EVOLUTIONARY OPTIMIZATION IN QUANTUM ENERGY TRANSPORT

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Abstract. We identify a critical complexity threshold $N_c \approx 5$ for quantum energy funnels. Below N_c, reliable funnels are trivially accessible. A narrow *Goldilocks zone* ($N = 5$–8) permits directed evolutionary search in $O(20)$ generations. Beyond $N \approx 9$, no combinatorial shortcut — symmetry, hierarchy, modularity — recovers tractability.

The Fenna–Matthews–Olson complex ($N = 7$) sits precisely at the evolutionary accessibility boundary. LH2 ($N = 27$) required precision band engineering, vibronic coupling, and billion-year timescales.

We establish this threshold by solving 50,000+ random Ising Hamiltonians across 14 system sizes ($N = 4$ to 30) using a 14-solver parallel optimization engine. The reliability curve follows $R(N) = 1/(1 + e^{0.50(N-5.0)})$ with fit error 5.5×10^{-4}. No local spectral feature predicts funnel reliability (all $|r| < 0.07$), proving the optimization is irreducibly global. At $N = 27$, no tested strategy produces reliable funnels: not C_9 symmetry (0.1%), not hierarchical dimer construction (0.0%), not modular assembly (0.0%).

These findings explain why photosynthetic quantum coherence evolved at specific scales and predict hard limits for synthetic biology approaches to artificial light harvesting.

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1. Introduction

Quantum coherence in photosynthetic energy transfer, first observed by Engel et al. [1] in the Fenna–Matthews–Olson (FMO) complex at 77 K and later confirmed at biological temperatures by Panitchayangkoon et al. [2], poses a fundamental question: why does quantum coherence survive thermal noise at 300 K?

Previous explanations focus on the *dynamics*: environment-assisted quantum transport [3], vibronic resonances [4], and noise-assisted efficiency [5]. We propose a complementary explanation from the *design* side: the FMO complex has exactly the right number of chromophores to be accessible to evolutionary optimization, and this number is determined by a sharp complexity threshold.

2. The Reliability Metric

Definition 2.1 (Funnel reliability)**.** For an N-site coupling Hamiltonian H with coupling matrix $J \in \mathbb{R}^{N\times N}_{\text{sym}}$, the *funnel reliability* $R(H)$ is the fraction of independent optimization solvers that find the same ground-state energy (within tolerance ϵ). A Hamiltonian is a *reliable funnel* if $R(H) > 0.8$.

We measure $R(H)$ using 14 parallel solvers (simulated bifurcation, coherent Ising, Kuramoto, simulated annealing, and 10 others) in the Isomorphic Engine [7]. When $R(H) > 0.8$, the energy landscape has a single dominant basin — a "funnel" that all solvers find regardless of initialization.

3. The Complexity Threshold

Theorem 3.1 (Complexity threshold)**.** *The fraction of random N-site symmetric Ising Hamiltonians that produce reliable funnels follows:*

$$R(N) = \frac{1}{1 + e^{0.50\,(N-5.0)}}, \qquad \textit{fit error} = 5.5 \times 10^{-4}, \tag{1}$$

with critical threshold $N_c = 5.0$ (the N at which $R = 50\%$).

Proof. Computed from 50,000+ random symmetric coupling matrices across 14 system sizes ($N = 4, 5, 6, 7, 8, 9, 10, 12, 15, 18, 20, 25, 27, 30$), each solved by 14 independent optimization algorithms. □

N	Parameters	Reliable (%)	Biological system
4	6	60.5	—
5	10	**50.4**	N_c (threshold)
6	15	38.8	—
7	21	**26.7**	FMO
8	28	18.5	PE545
10	45	8.0	—
15	105	1.2	LHCII
20	190	0.3	—
27	351	0.0	LH2

4. The Goldilocks Zone

Proposition 4.1 (Directed evolution in the Goldilocks zone)**.** *At $N = 7$, a $(1 + 1)$ evolutionary strategy with mutation rate $\sigma = 0.1$ finds a reliable funnel in 21 ± 35 generations (mean $\pm$ std over 50 independent trajectories, 100% success rate).*

Proof. Each trajectory starts from a random 21-parameter genome (the off-diagonal elements of J), mutates by adding $\mathcal{N}(0, \sigma)$ to each coupling, evaluates funnel reliability, and selects the better of parent and child. All 50 trajectories found $R > 0.8$ within 5,000 generations (all within 181). □

The Goldilocks zone $N \in [5, 8]$ is characterized by:

- Random success rate 18–50%: non-trivial but not hopeless.
- Directed evolution succeeds in $O(10)$ generations: tractable on evolutionary timescales.
- Below $N = 5$: too simple for complex energy routing.
- Above $N = 8$: reliability drops below 18%, requiring exponentially more search time.

5. Beyond the Threshold: Why LH2 is Hard

Theorem 5.1 (No combinatorial shortcut for $N \geq 18$)**.** *At $N = 27$ (the LH2 scale), no tested combinatorial strategy produces reliable funnels:*

Strategy	*Reliable (%)*	*Free parameters*
Random (no structure)	*0.0*	*351*
C_9 symmetric	*0.1*	*4*
C_9 with fixed dimers	*0.0*	*2*
C_3 symmetric	*0.1*	*∼40*
Hierarchical (dimer ring)	*0.0*	*5*

Corollary 5.2. *The LH2 light-harvesting complex required optimization channels beyond combinatorial search:*

(i) ***Band engineering****: specific coupling signs and magnitudes creating directional energy gradients (not achievable by random sampling).*

(ii) ***Vibronic coupling****: protein vibrations providing* $\sim 35\,\mathrm{cm}^{-1}$ *reorganization energy as an additional optimization channel.*
(iii) ***Geological timescales****:* ~ 3 *billion years of directed evolution with small mutation steps.*
(iv) ***Modular bootstrapping****: evolving FMO-scale units first, then assembling them with precision engineering.*

6. Spectral Unpredictability

Proposition 6.1 (No spectral shortcut)**.** *No local spectral feature of the coupling matrix (frustration ratio, density, coupling standard deviation, coupling max/min ratio) predicts funnel reliability. All Pearson correlations* $|r| < 0.07$ *over 5,000 random* $N = 7$ *Hamiltonians.*

This connects to computational complexity: if a polynomial-time spectral predictor existed, it would provide a fast oracle for a hard optimization class. The absence of correlation suggests funnel-finding is *irreducibly global* — the landscape must be traversed to be evaluated.

7. Discussion

7.1. **Why photosynthesis stopped at seven.** The FMO complex has 7 bacteriochlorophyll sites not because 7 is "special" in any group-theoretic sense, but because 7 is the **largest system size accessible to evolutionary optimization** without additional physical channels. At $N = 7$:

- 26.7% of random Hamiltonians produce reliable funnels.
- Directed evolution finds one in ~ 21 generations.
- The optimization is tractable on evolutionary timescales ($\sim 10^6$ years per generation at mutation rates of $\sim 10^{-3}$ per gene per generation for proteins).

At $N = 15$ (LHCII scale), reliability drops to 1.2%, requiring $\sim 80\times$ more search. At $N = 27$ (LH2), random search is futile — evolution needed vibronic coupling, band engineering, and billions of years.

7.2. **Falsifiable predictions.**

P1. Photosynthetic complexes with $N > 10$ independent (non-symmetric) chromophores and reliable energy funnels should not exist in nature. If found, the threshold model is wrong.
P2. *In vitro* directed evolution of small chromophore arrays should find efficient energy transfer at $N \leq 8$ within ~ 100 rounds of selection, but fail at $N \geq 15$ without vibronic assistance.
P3. The critical threshold N_c should depend on the coupling distribution: systems with sparser, more structured couplings (e.g., chain vs. all-to-all) should have higher N_c.

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